

Title: Smart Home Finder with Real-World Insights

WEBSITE OVERVIEW

The **House-Hunting App** is an intelligent digital platform designed to help users easily search and discover properties based on their preferences. The system provides property listings, location details, budget filters, and amenities information to help users find suitable homes quickly. The platform acts as a smart guide for buyers and renters by displaying property comparisons, nearby facilities, and personalized suggestions. By analyzing user preferences, search behavior, and property availability, the system provides a smooth and efficient house-search experience, making property discovery easier and more convenient.

MARKET RELEVANCE

As more people prefer online platforms to search for homes, property-search applications have become highly important. Users expect **fast search results, clear property details, and easy comparison options**.

The House-Hunting App provides:

- Quick property search based on location
- Budget-based filtering
- Amenities and facility details
- Easy comparison between properties
- Personalized suggestions for suitable homes

Within the system, the recommendation and search features improve the overall property discovery experience and help users make faster decisions.

PROBLEM IT SOLVES

Traditional house-search methods cause users to:

- Spend too much time browsing multiple websites

- Struggle to find properties within their budget
- Face difficulty comparing multiple houses
- Lack clear information about nearby amenities
- Feel confused when too many property options are available

The House-Hunting App solves these problems by providing **organized property listings, filtering options, and comparison tools** in a single platform.

INTRODUCTION

PROBLEM STATEMENT

People searching for houses often face difficulty finding properties that match their **location preferences, budget limits, and required facilities**. Existing platforms may provide many property listings but lack personalized suggestions and easy comparison features. Users often struggle to understand property details, neighborhood information, and nearby amenities. This leads to confusion and delays in decision-making.

PURPOSE

The purpose of this system is to create a **centralized house-search platform** that allows users to easily explore available properties and filter them based on their needs. The system helps users compare houses, view property details, and make informed decisions quickly.

SCOPE

The House-Hunting App supports:

- Property search based on location
- Budget-based property filtering
- Viewing property details and images
- Amenities and nearby facilities information
- Property comparison features
- Saving favorite properties

- Personalized property recommendations

The system focuses on improving the property search experience and simplifying decision-making for users.

GPCU FRAMEWORK

G – GAP ANALYSIS

Current real estate platforms provide property listings but lack personalized recommendations and advanced filtering features. Users often struggle to quickly find suitable homes based on their preferences.

The House-Hunting App fills this gap by providing **smart filtering, personalized suggestions, and easy property comparison.**

P – PRODUCT

The product is a **House-Hunting Web Application** that allows users to search and explore properties based on location, budget, and amenities. The platform provides property details, images, and comparison tools to help users choose the best home.

C – COMPETITION ANALYSIS

Existing platforms such as **MagicBricks**, **99acres**, and **Housing.com** offer property listings and search filters but mainly focus on basic property browsing features.

U – UNIQUENESS

- The proposed system provides:
 - Smart property filtering based on user preferences.
 - Easy comparison of multiple houses.
 - Location-based property discovery.
 - Personalized property recommendations.
 - Simple and user-friendly interface.

This makes the platform more efficient and convenient for users searching for homes.

FUTURE POTENTIAL

- AI-based property recommendation systems.
- Virtual house tours using 3D models.
- Real-time neighborhood safety information.
- Integration with property loan and EMI calculators.
- Mobile app expansion with location-based alerts.

WHY USERS CONTINUE USING IT

- Quick and easy property search.
- Clear property details and images.
- Simple filtering based on budget and location.
- Personalized recommendations.
- Convenient property comparison features.

The House-Hunting App becomes a **complete platform for property discovery and decision-making**.

DESIGN & ENGINEERING STANDARDS — Chat Agent

User-Centered Design

- Simple and user-friendly interface.
- Easy property search options.
- Clear property details and images.
- Accessible design for all users.

Software Engineering Practices

- Database management for property listings.
- Secure login and authentication systems.
- Efficient search algorithms for property filtering.
- Fast backend processing for quick search results.

Data Security & Privacy

- Secure user authentication.
- Protection of personal user information.
- Secure storage of property data.
- Users can manage their personal information.

Performance Standards

Fast property search results

Efficient system performance during high usage

Quick loading of property images and details

Testing & Quality Assurance

- Functional testing for property search features
- Performance testing for system speed
- Usability testing with real users
- Error handling and system reliability testing

Interoperability Standards

The system supports integration with:

- Property listing databases
- Location and map services
- Real estate data platforms
- Payment gateways for booking services.

Ethical & User Transparency Standards

- Clear property details and pricing information.
- Transparent listing of property features and amenities.
- Ethical handling of user data.
- User control over personal information and saved properties.

Relevant Standards for Real-Estate Platforms

Data Privacy & Security Standards

ISO/IEC 27001 — Secure handling of user data and property information.

Software Quality Standards

ISO/IEC 25010 — Ensures system usability, reliability, and performance.

Usability & Accessibility Standards

WCAG 2.2 Accessibility Guidelines:

- Readable text and simple navigation

- Screen-reader compatibility
- Accessible design for different users

FUNCTIONAL PROTOTYPE

Key Functionalities Implemented

User Authentication & Profiles

- Secure user signup and login
- User profiles storing preferences and saved properties
- Users can update their profile information

Property Search Module

- Users can search houses based on location
- Filter properties based on budget and amenities
- Display property images and descriptions

Property Comparison Feature

- Compare multiple houses side-by-side
- Show differences in price, amenities, and location
- Helps users make better decisions

Property Recommendation System

- Suggests houses based on user preferences
- Shows similar properties related to search results

- Improves property discovery experience

Favorite Properties

- Users can save properties for later viewing
- Easy access to shortlisted houses
- Helps users track preferred properties

Property Details & Viewing

- Shows property images and descriptions
- Displays location information and nearby facilities
- Helps users understand property features clearly

Payment Integration

- Supports **UPI, card, wallet, and online payment options** for booking property visits or reservation fees.
- Ensures **secure transaction flow through encrypted APIs** for safe payment processing.

Feedback & Review System

- Users can submit property reviews directly through the platform.
- The system collects feedback regarding property quality, neighborhood experience, and overall satisfaction.
- Helps improve future property recommendations based on user feedback.

History & Interaction Records

- Search history, property views, and previous interactions are stored (with user permission).
- Enables the system to provide more personalized property recommendations in future searches.
- Browsing history and saved properties help refine AI-based suggestions.

Utility & Settings

- Users can clear search history, manage profile details, and adjust notification settings.
- Option to delete account and stored data in compliance with privacy standards.

Prototype Testing & Validation

- Full UI-UX testing for smooth navigation across all screens.
- Tested end-to-end flow:
Splash → Login → Home → Property Search → Property Details → Comparison → Booking/Contact
- API integration validated for real-time property listings, price updates, and availability information.
- System tested for:
 - Search accuracy
 - Property recommendation quality
 - Filter functionality
 - Property comparison clarity
- Ensured compatibility across devices, browsers, and web environments.
- Stress-tested for handling multiple users and large property datasets.

User Feedback Summary

- Test users appreciated the **simple and fast property search process**.
- Property suggestions based on **location and budget improved decision making**.
- The **property comparison feature** was considered very useful.
- Users found the **filter-based search easier than browsing multiple websites**.
- Overall, the house-hunting experience felt **more organized, efficient, and user-friendly**.

CONCLUSION

The **House-Hunting App transforms traditional property searching into a smart and convenient digital experience**. By combining organized property listings, intelligent filtering, personalized recommendations, and easy property comparison, the platform simplifies the process of finding suitable homes. The system helps users quickly discover properties that match their **location, budget, and amenities**, making house searching faster, clearer, and more efficient.